

```

%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path,'sym4D_cutsAndFits');

if ~exist('cuts2fit200GP','var')
    ld = load(fullfile(data_path,'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200GP','var')
    used_ds = used_ds200GP;
else
    used_ds = containers.Map()
    sel_info = struct('cuts',[],'en_range',[],'fname','', 'bg_par',[]);
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off110f};
    sel_info.en_range = 30:10:120;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off110_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir111')=sel_info;
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off200f};
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off200_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir111')=sel_info;
    %
    used_ds200GP = used_ds;
end

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
    KeyType: char
    ValueType: any

```

```

select= 'Ei200Br200dir111';
sel_info = used_ds(select);

cuts_list = sel_info.cuts

```

```

cuts_list = 1x1 cell array
    {1x1 sqw}

```

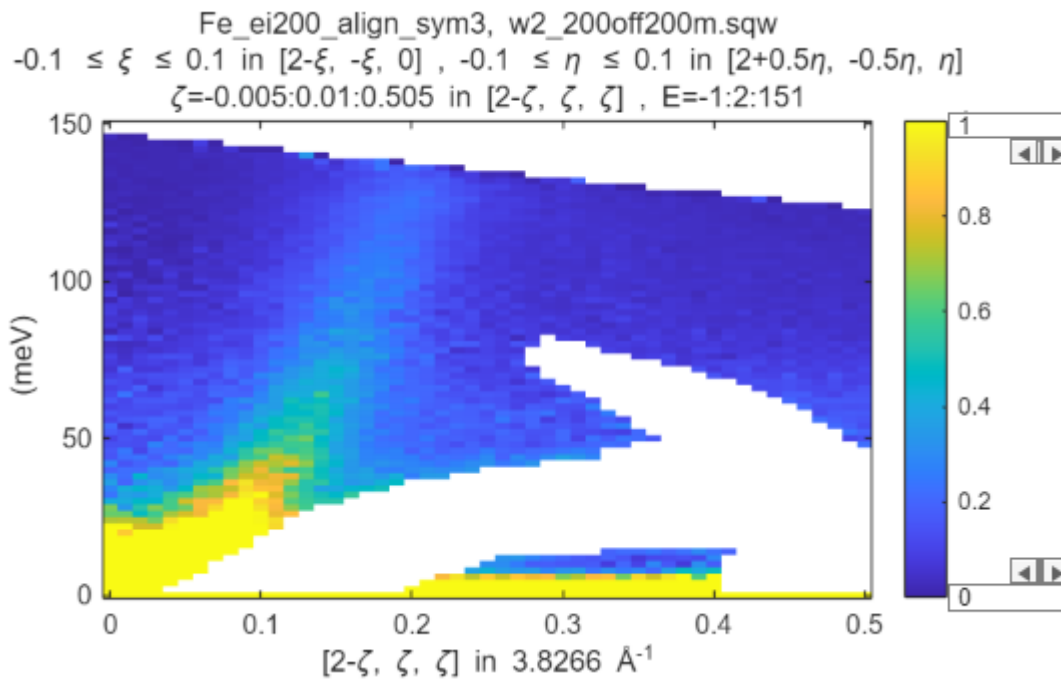
```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)

```

```
cuts_list{i} = cuts_list{i}.set_instrument(mi);
cuts_list{i} = cuts_list{i}.set_sample(sample);
plot(cuts_list{i}); lz 0 1; keep_figure;
```

end



```
dir_name = "GP";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in  
going finer
```

```
half_dE = 10; % half width of data binning
```

```
cut_en = sel_info.en_range;
```

```
cut_dimensions = sel_info.cut_dim;
```

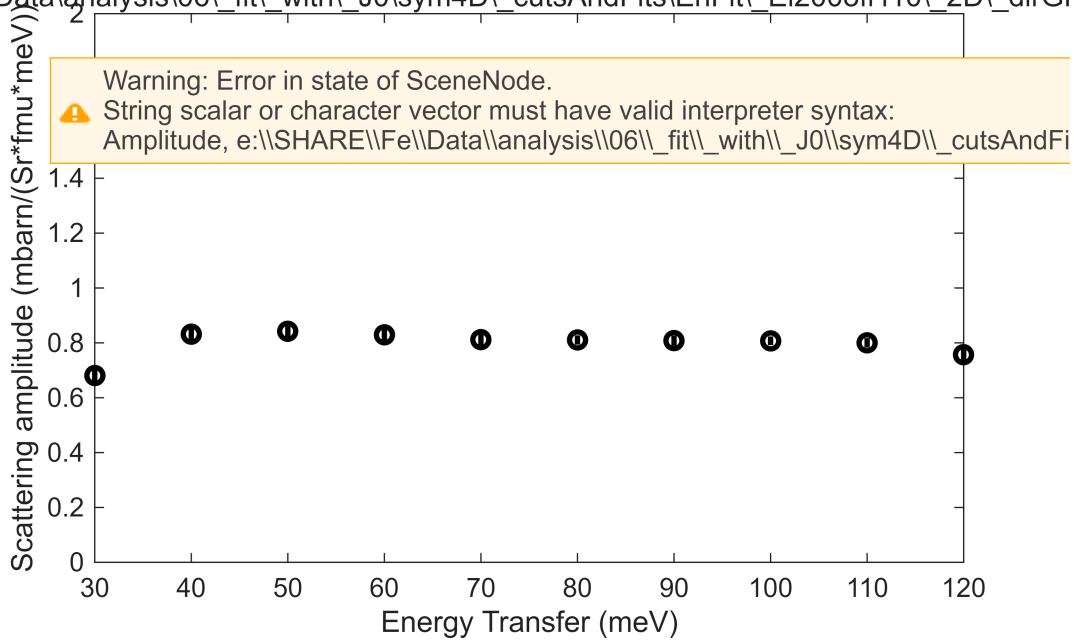
```
bg_par = sel_info.bg_par;
```

```
cuts_list{end+1} = bg_par;
```

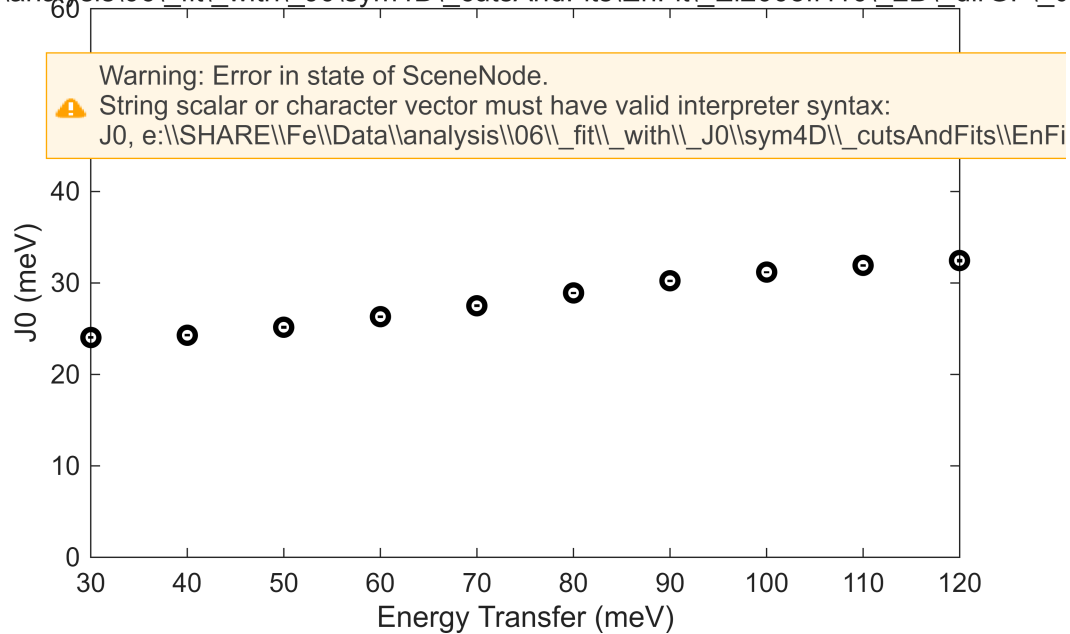
```
fbase_name = sel_info.fname;
```

```
[fit_res_rec,res_name] = fit_multicuts_along_direction(...  
cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

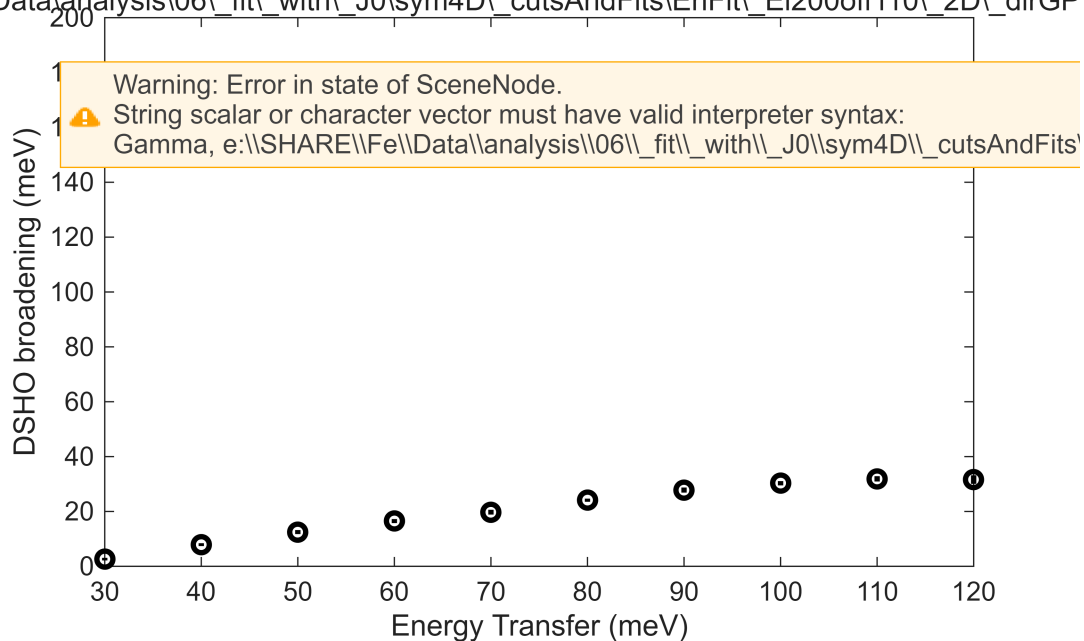
Fe\\Data\\analysis\\06_fit_with_J0\\sym4D_cutsAndFits\\EnFit_Ei200off110_2D_dirGf



Data\\analysis\\06_fit_with_J0\\sym4D_cutsAndFits\\EnFit_Ei200off110_2D_dirGP_c



\\e\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits\EnFit_Ei200off110_2D_dirGP



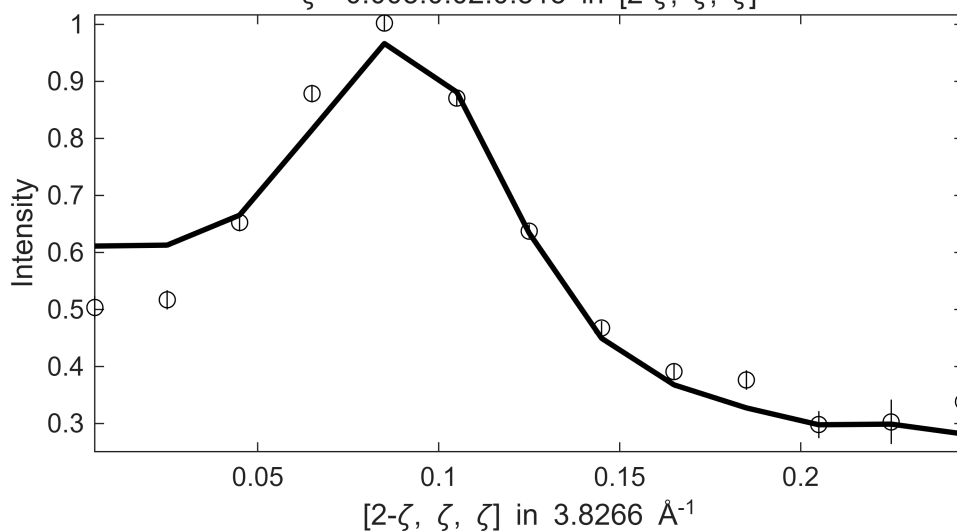
**** En = 30±10

Fe_ei200_align_sym3, w2_200off200m.sqw

S=0.69±0.03; J0= 23±0.33; gamma=3.08653±0.109358

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 20 ≤ E ≤

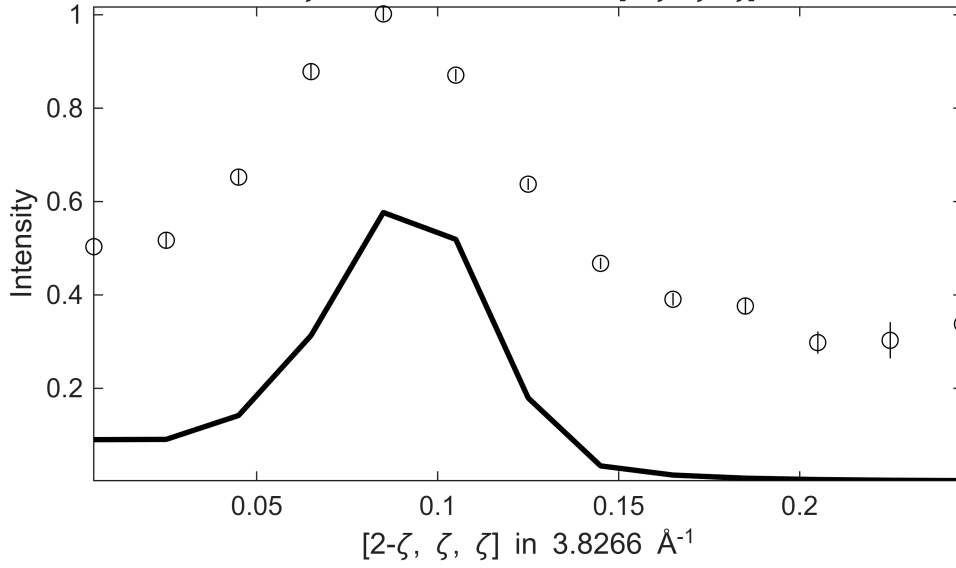
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $20 \leq E \leq$

$\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



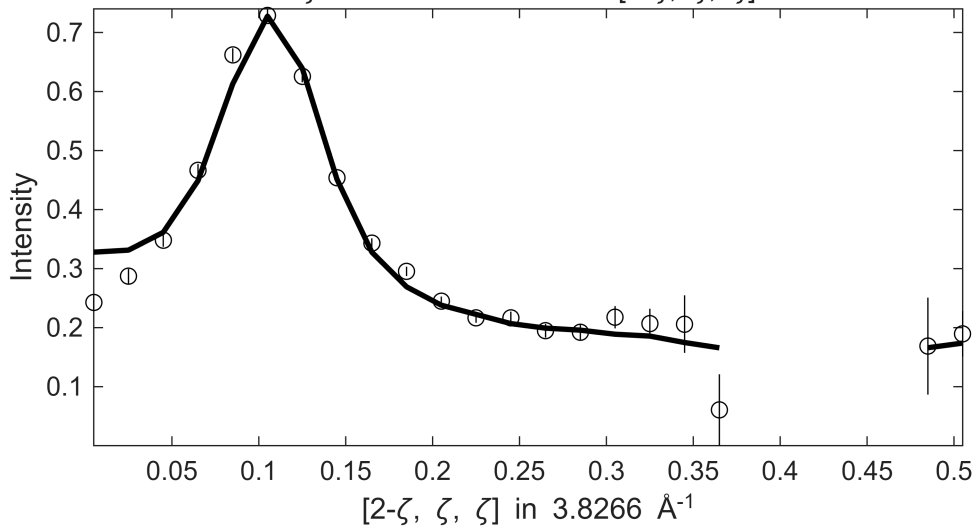
**** En = 40 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw

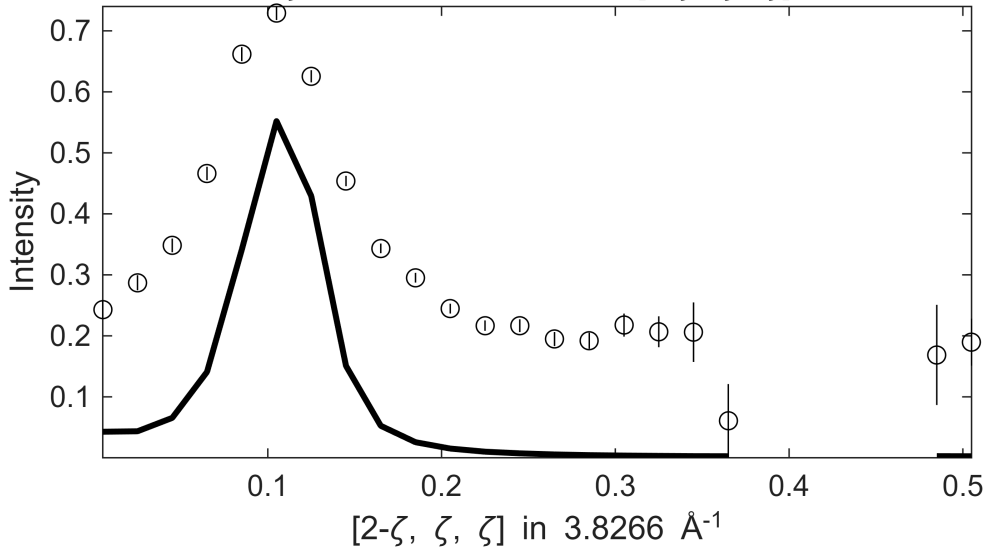
S= 0.84 ± 0.018 ; J0= 24 ± 0.21 ; gamma= 7.3314 ± 0.147052

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $30 \leq E \leq$

$\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

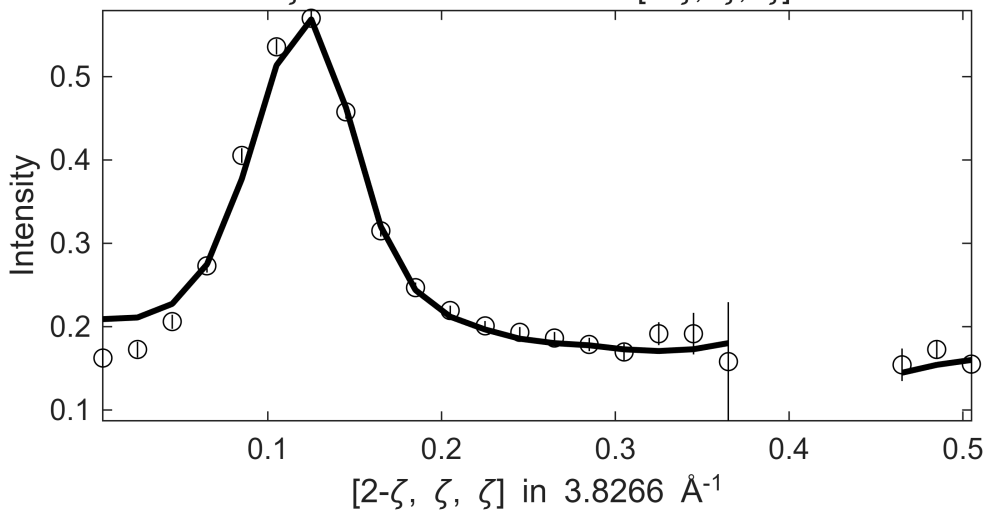


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $30 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

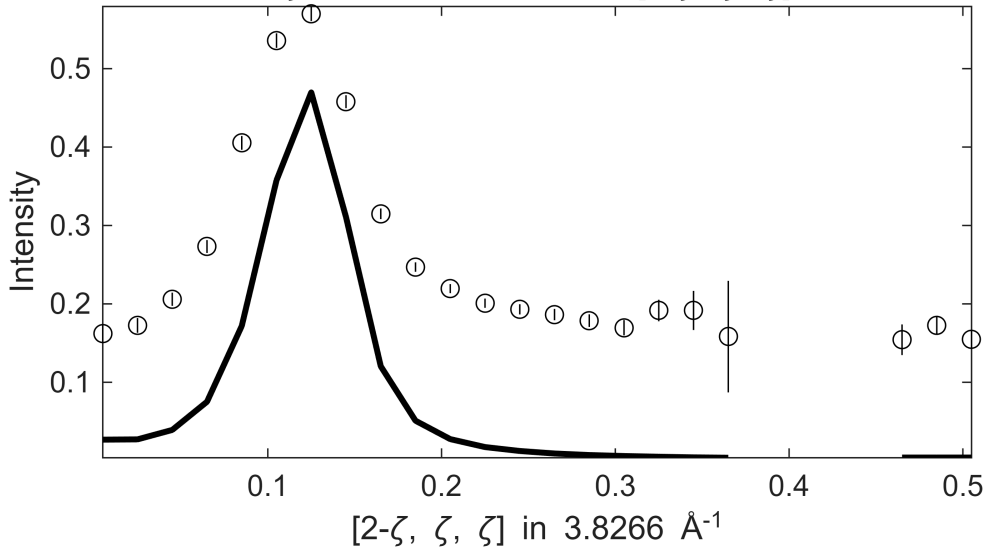


 **** En = 50 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S = 0.88 \pm 0.044$; $J_0 = 25 \pm 0.27$; $\gamma = 11.4408 \pm 1.12001$
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $40 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

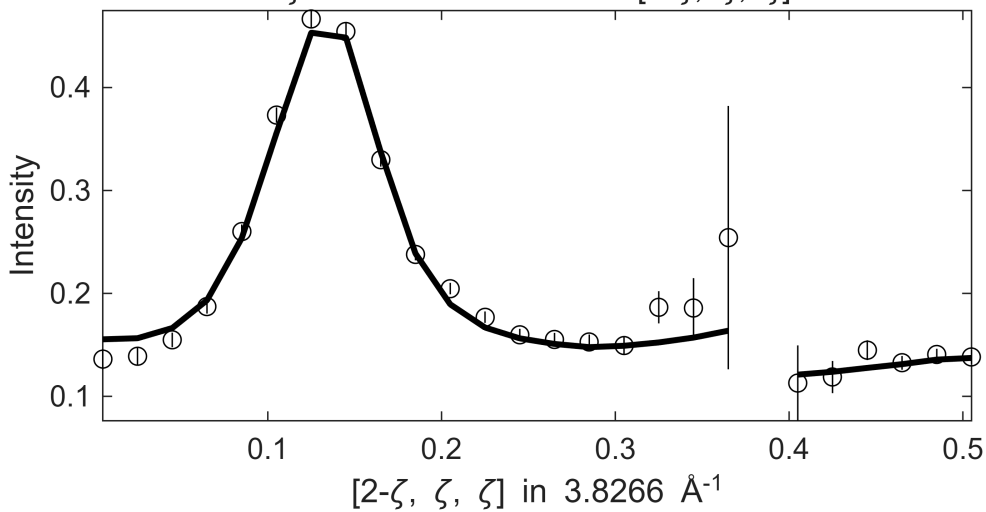


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $40 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

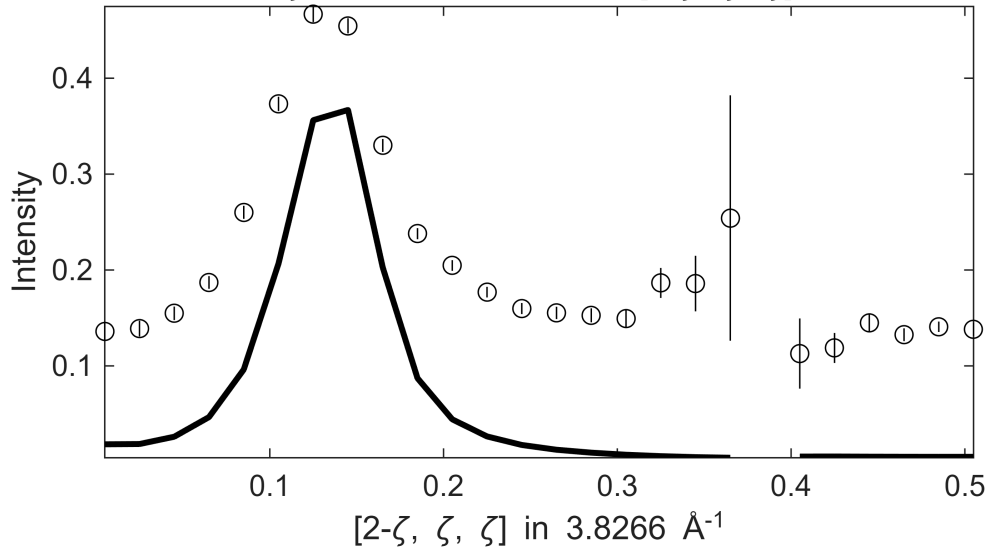


 **** En = 60±10

Fe_ei200_align_sym3, w2_200off200m.sqw
 S=0.86±0.014; J0= 26±0.17; gamma=15.0183±0.222944
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $50 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

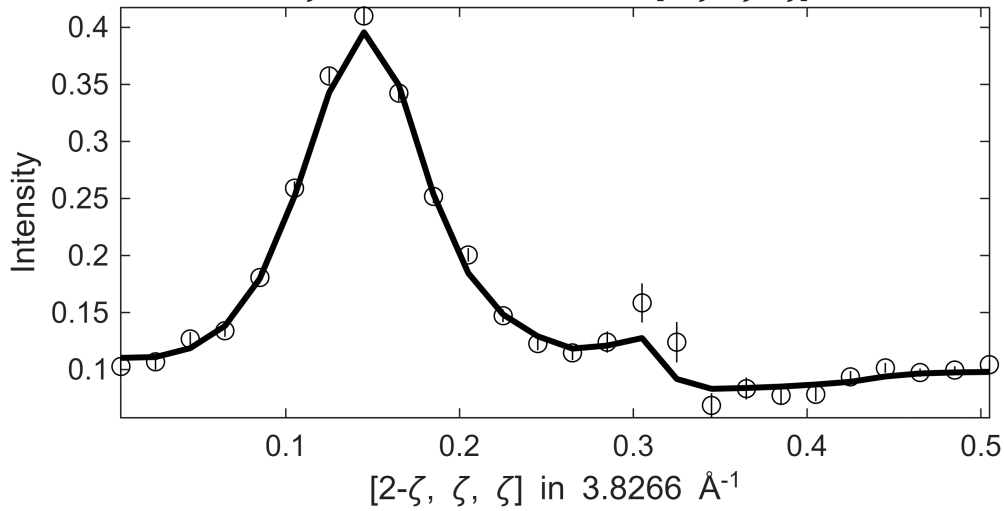


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $50 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

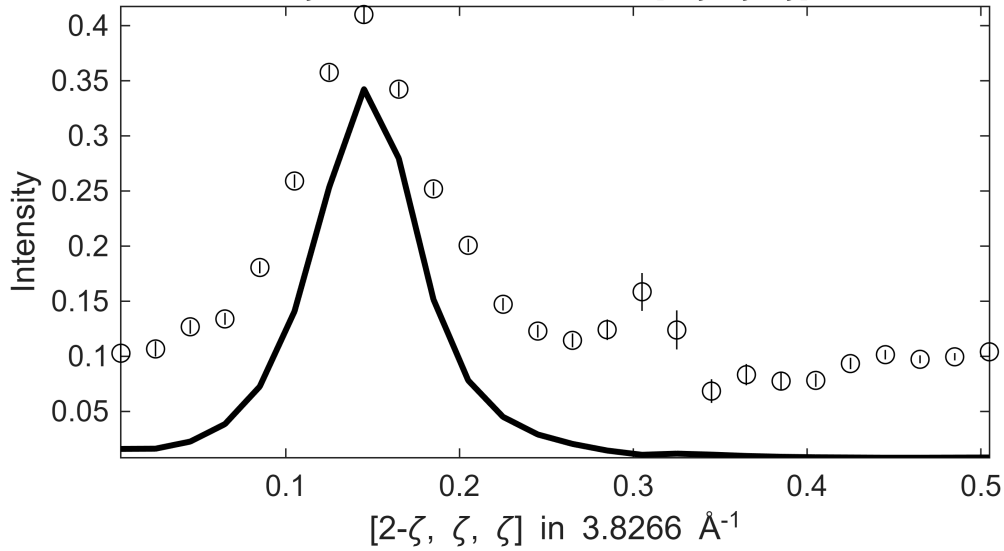


 **** En = 70 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S=0.95 \pm 0.038$; $J_0 = 28 \pm 0.25$; $\gamma = 21.0399 \pm 1.35893$
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $60 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

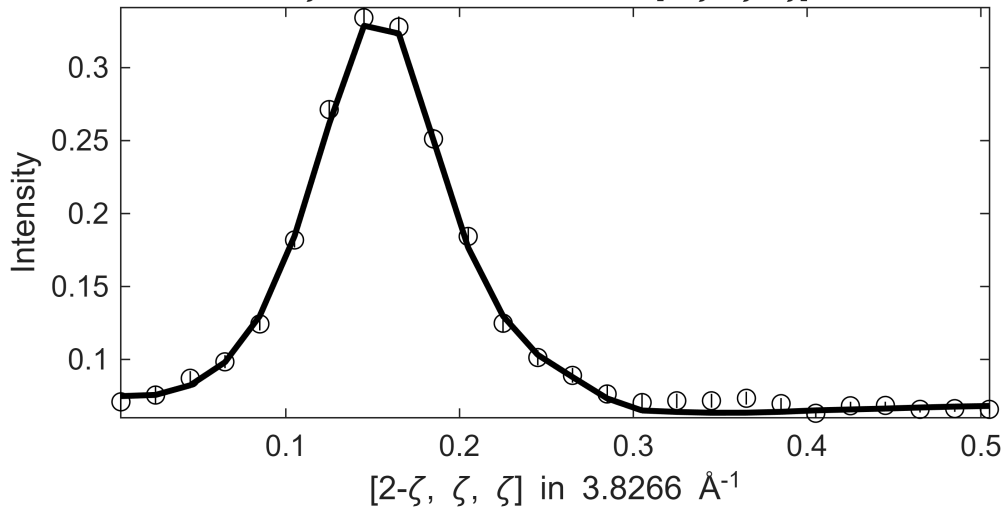


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $60 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

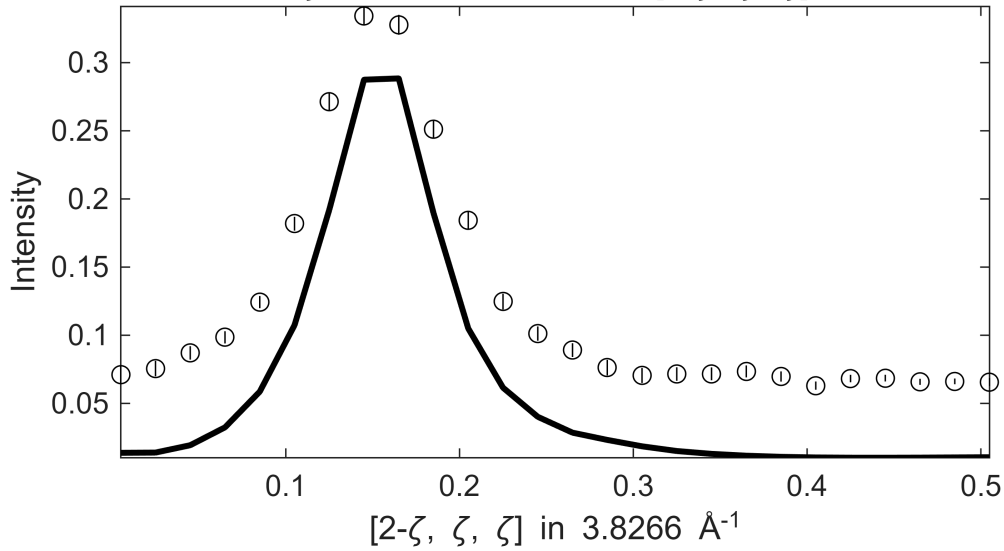


 **** En = 80 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 S= 1 ± 0.034 ; J0= 29 ± 0.26 ; gamma= 26.0795 ± 1.39845
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $70 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

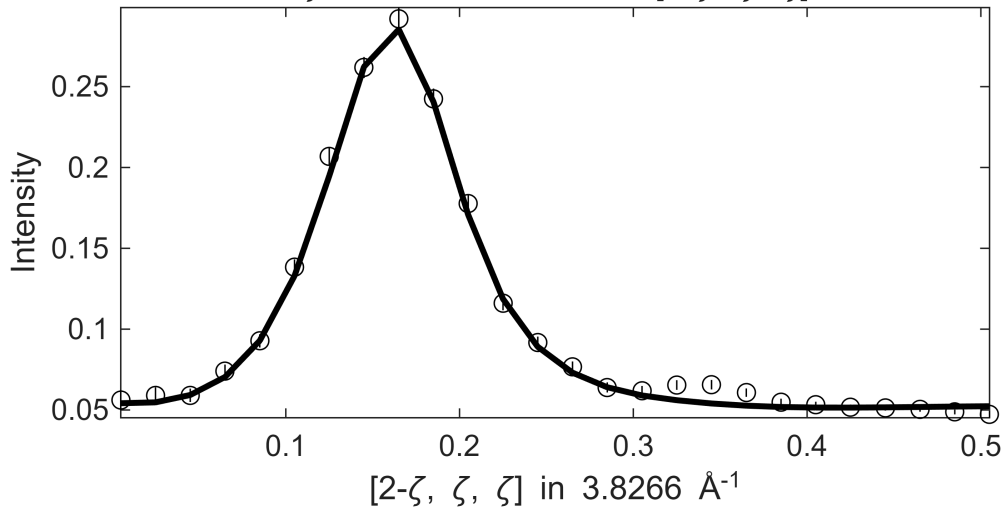


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $70 \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

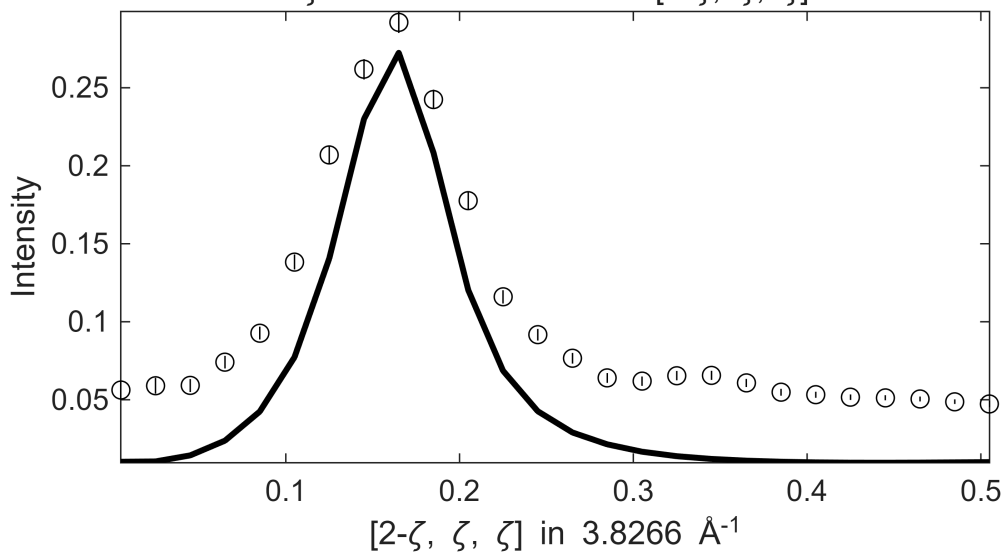


 **** En = 90 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S=0.91 \pm 0.032$; $J_0= 31 \pm 0.29$; $\gamma=27.6699 \pm 1.56655$
 $\leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $80 \leq$ ξ
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

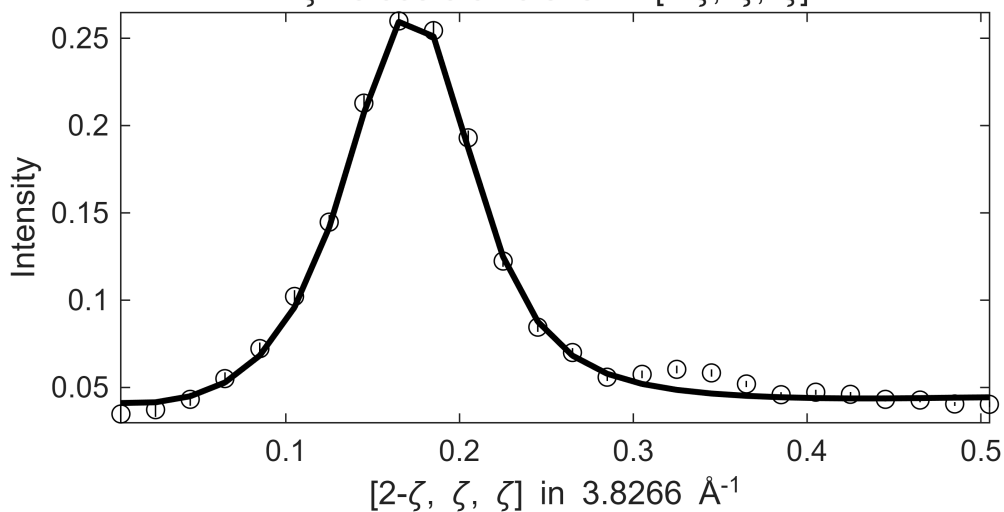


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $80 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

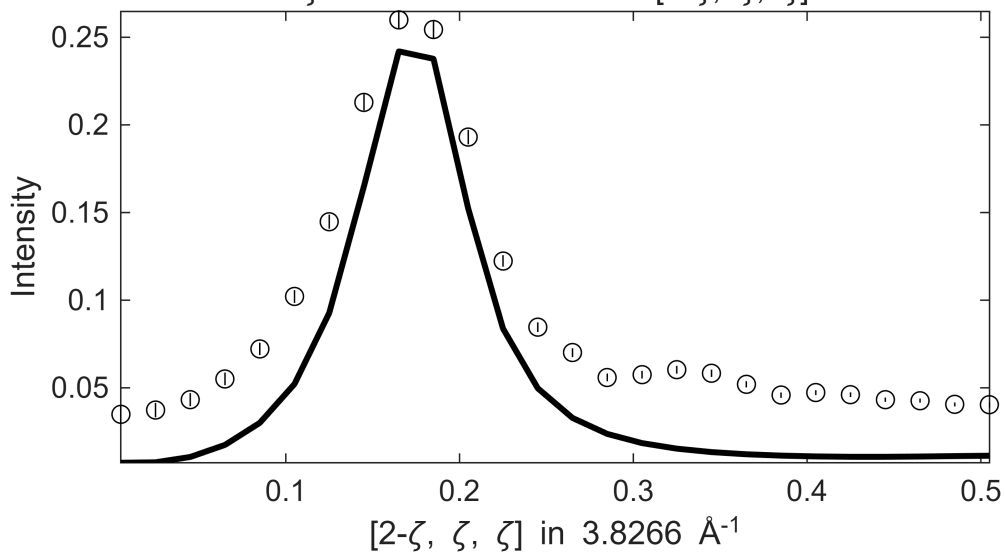


 **** En = 100 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S=0.87 \pm 0.028$; $J_0 = 32 \pm 0.25$; $\gamma = 29.0272 \pm 1.5198$
 $\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $90 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

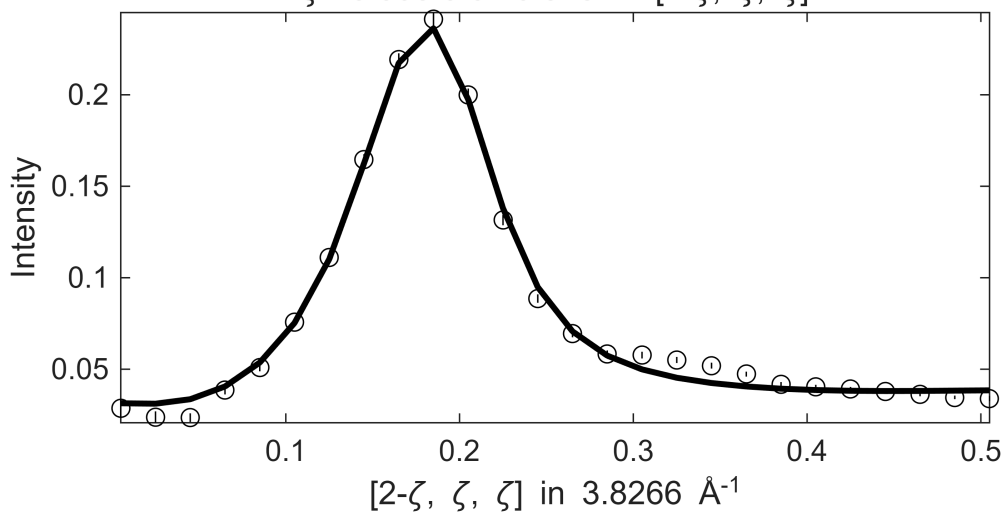


Fe_ei200_align_sym3, w2_200off200m.sqw
 $\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $90 \leq \theta$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



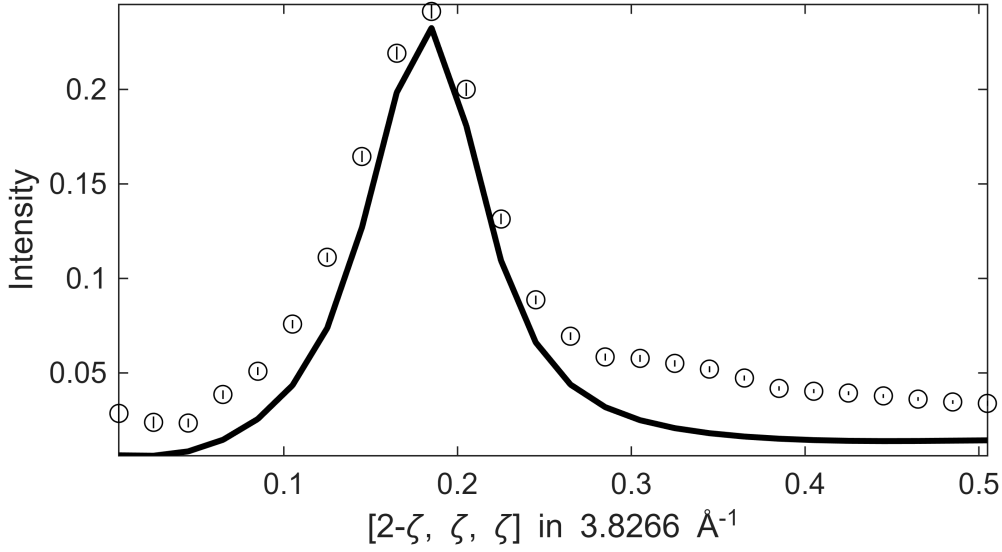
 **** En = 110 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S=0.91 \pm 0.027$; $J_0=33 \pm 0.23$; $\gamma=35.0355 \pm 1.59419$
 $\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $100 \leq \theta$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200m.sqw

$\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $100 \leq \zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



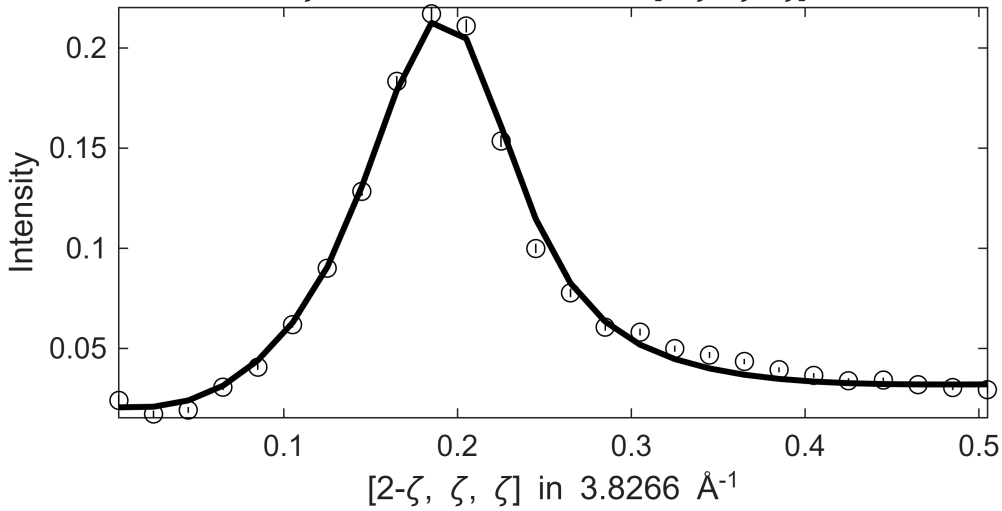
**** En = 120±10

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.3986

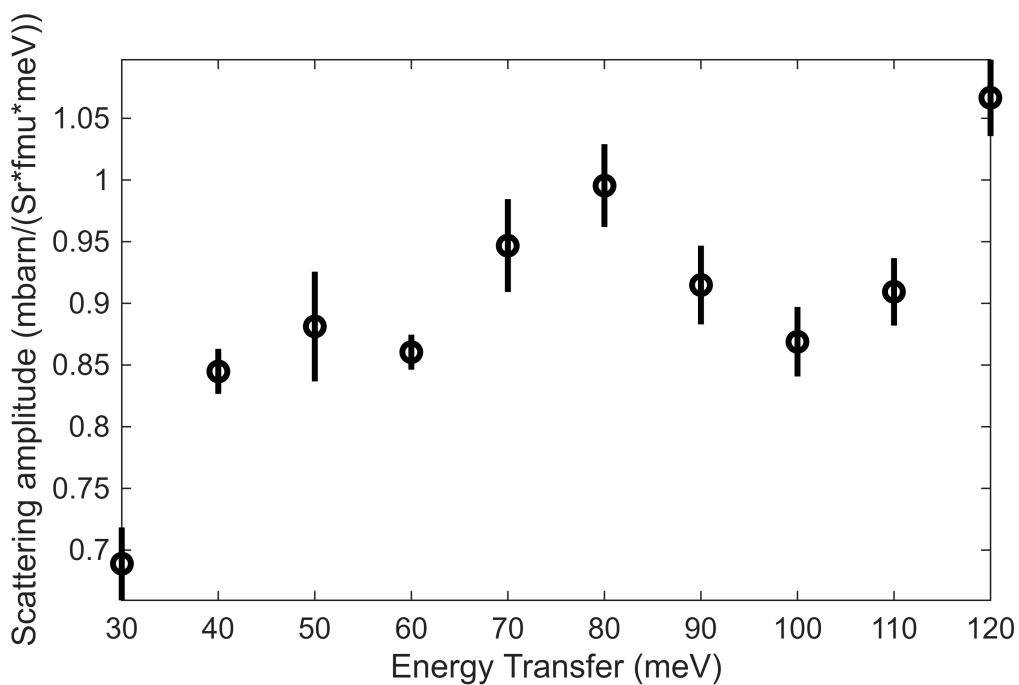
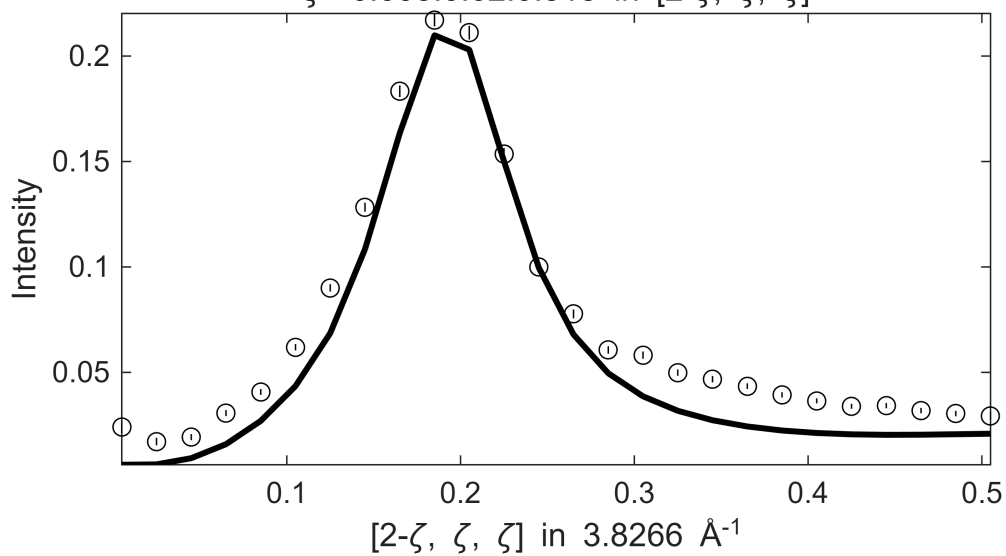
Fe_ei200_align_sym3, w2_200off200m.sqw

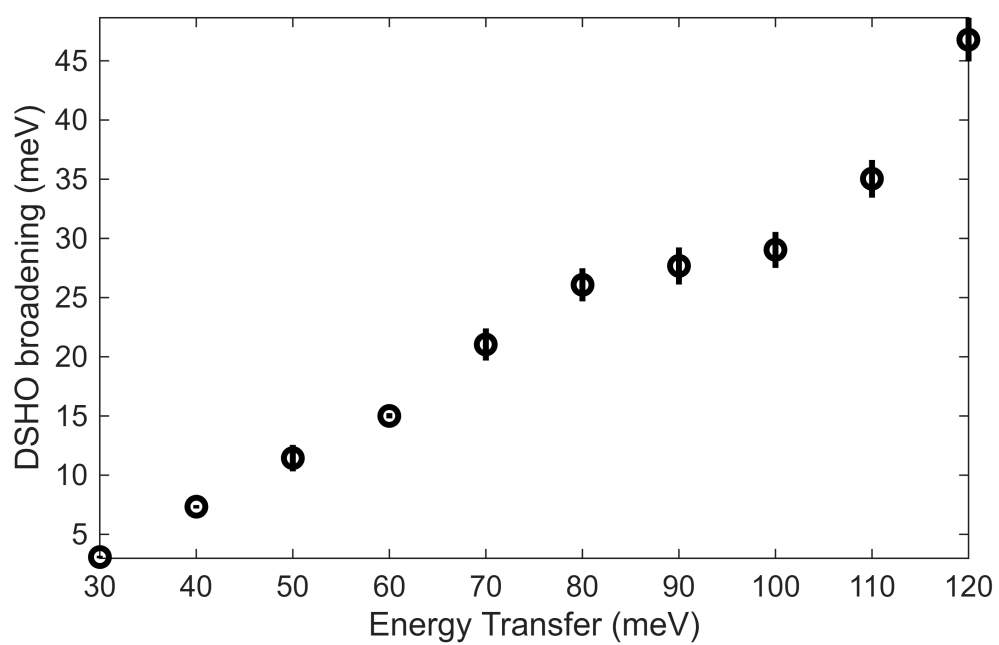
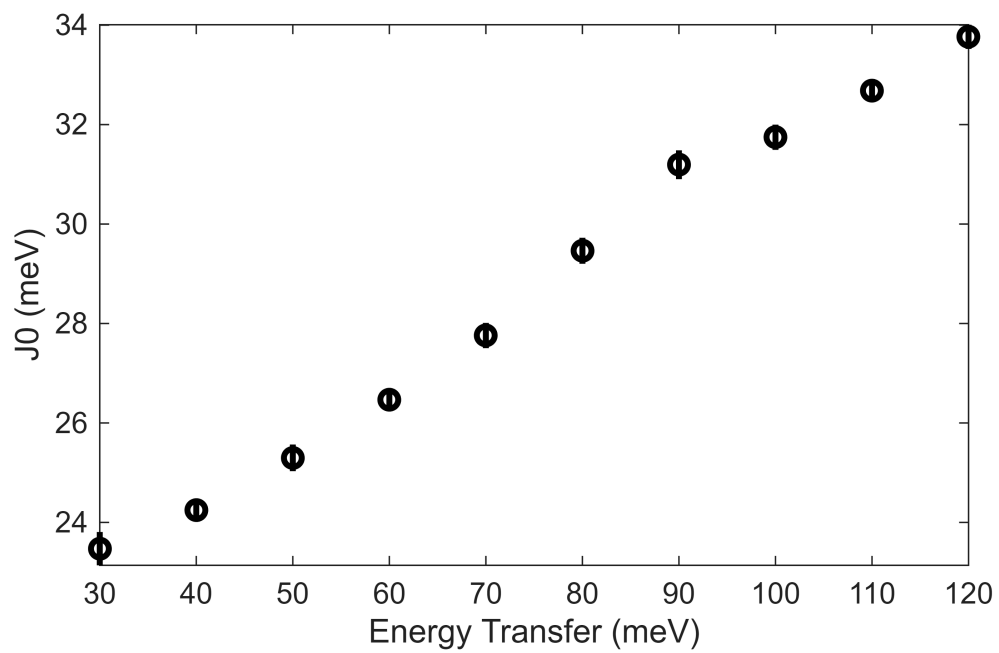
S= 1.1±0.031; J0= 34±0.24; gamma=46.7957±1.84392

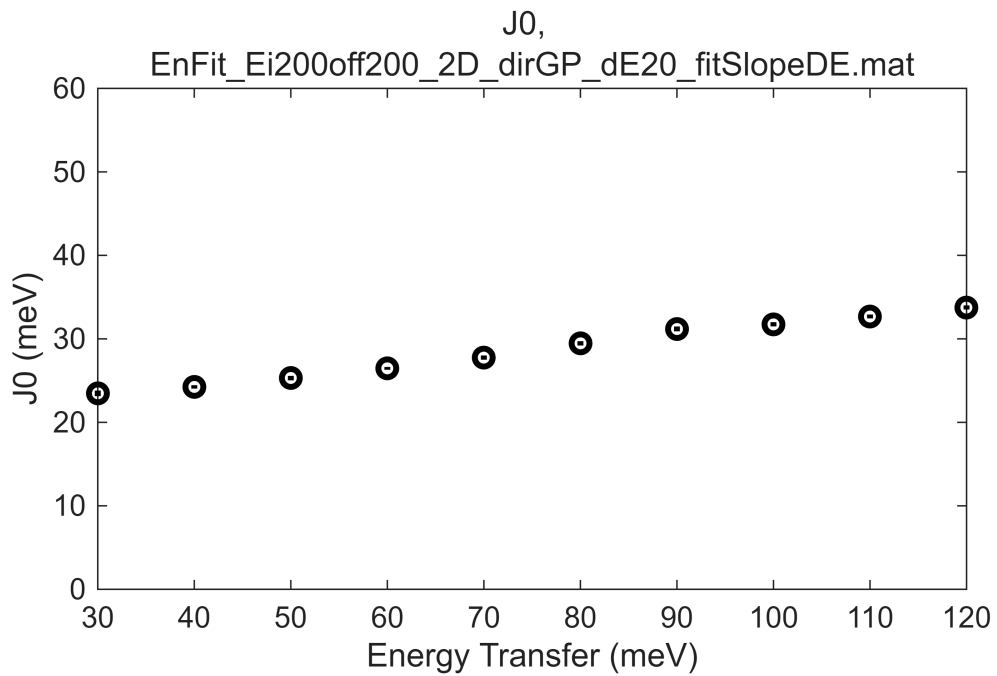
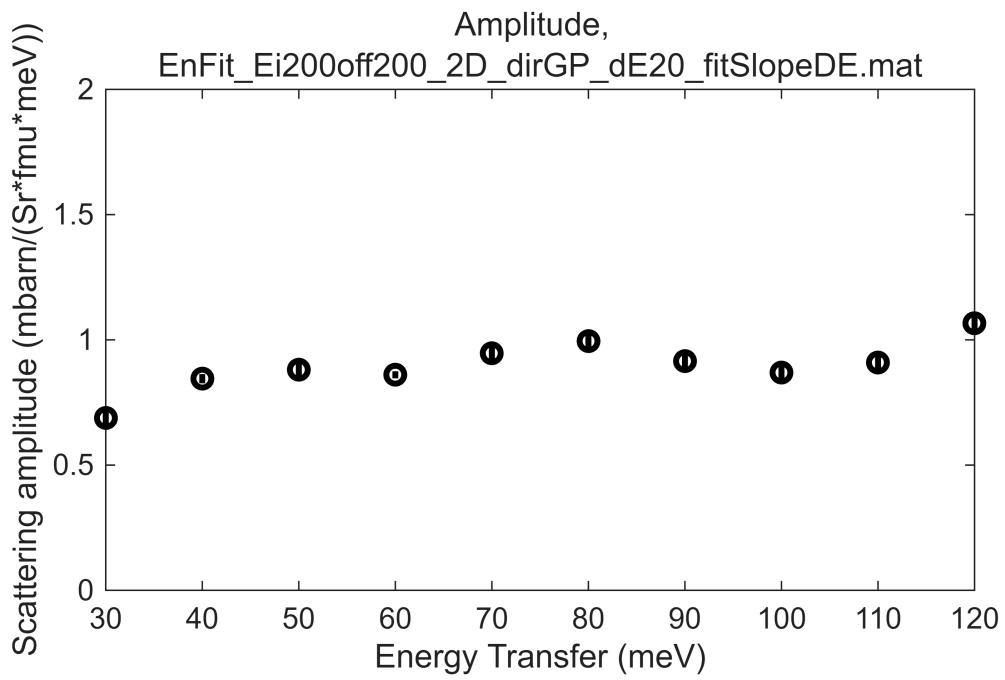
$\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $110 \leq \zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

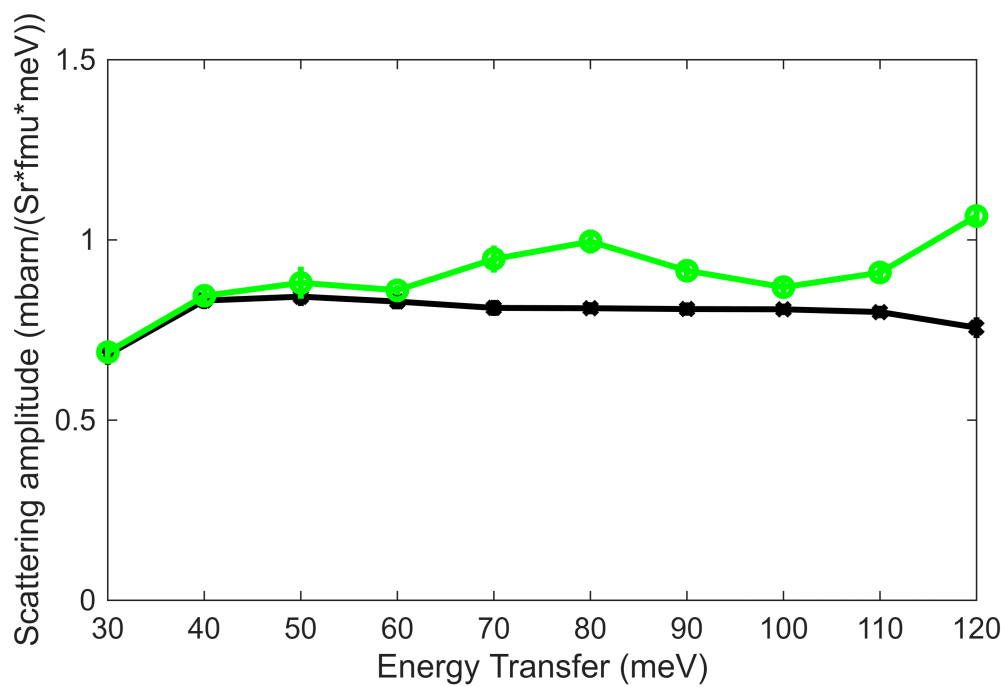
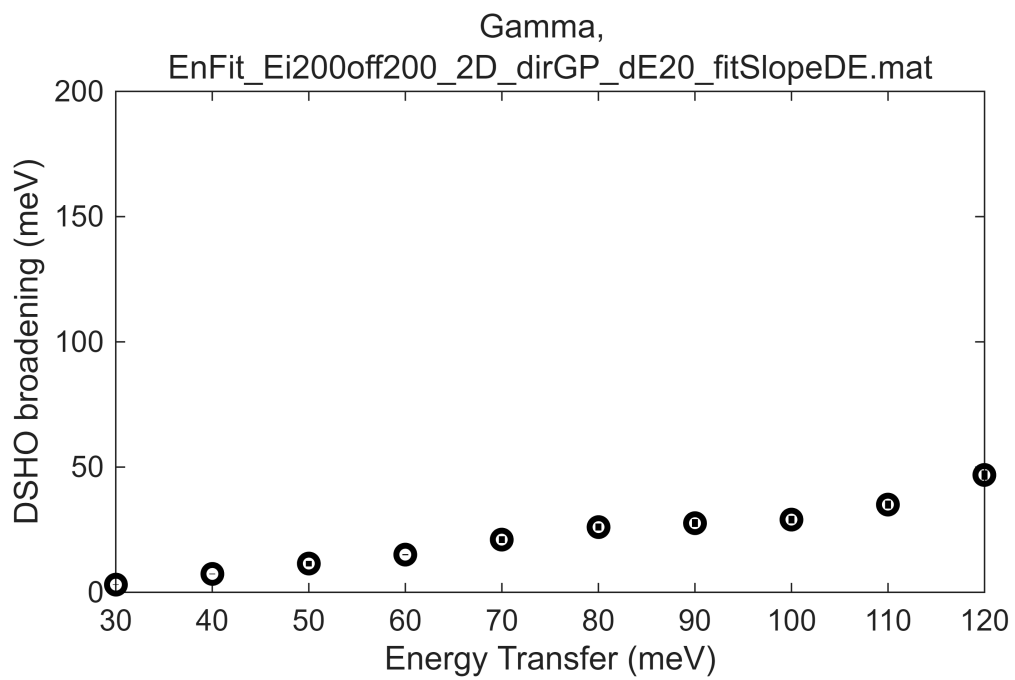


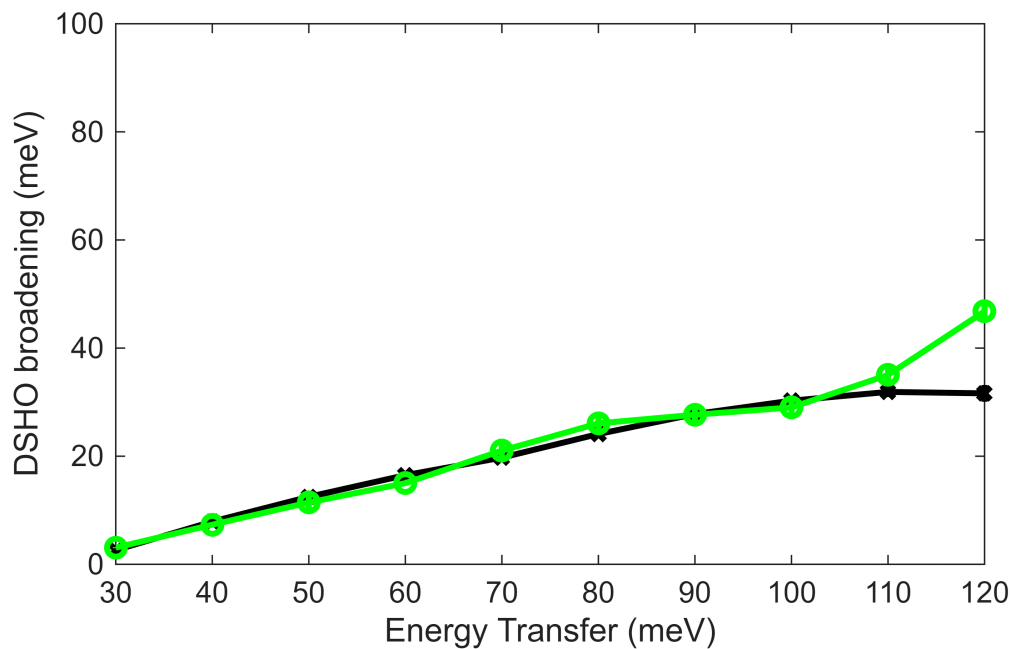
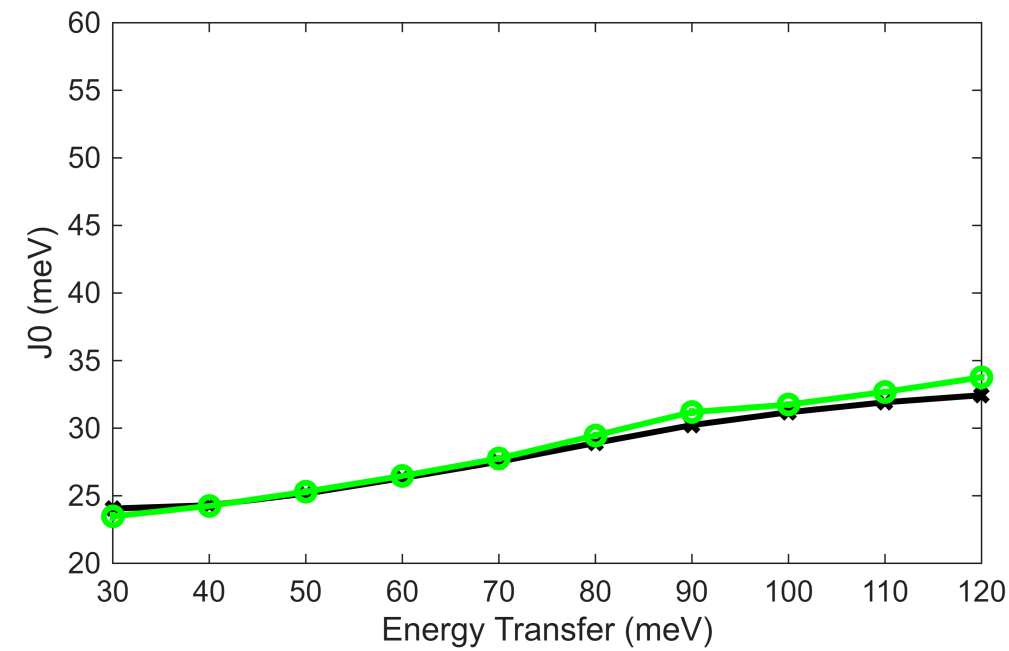
Fe_ei200_align_sym3, w2_200off200m.sqw
 $\xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $110 \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$











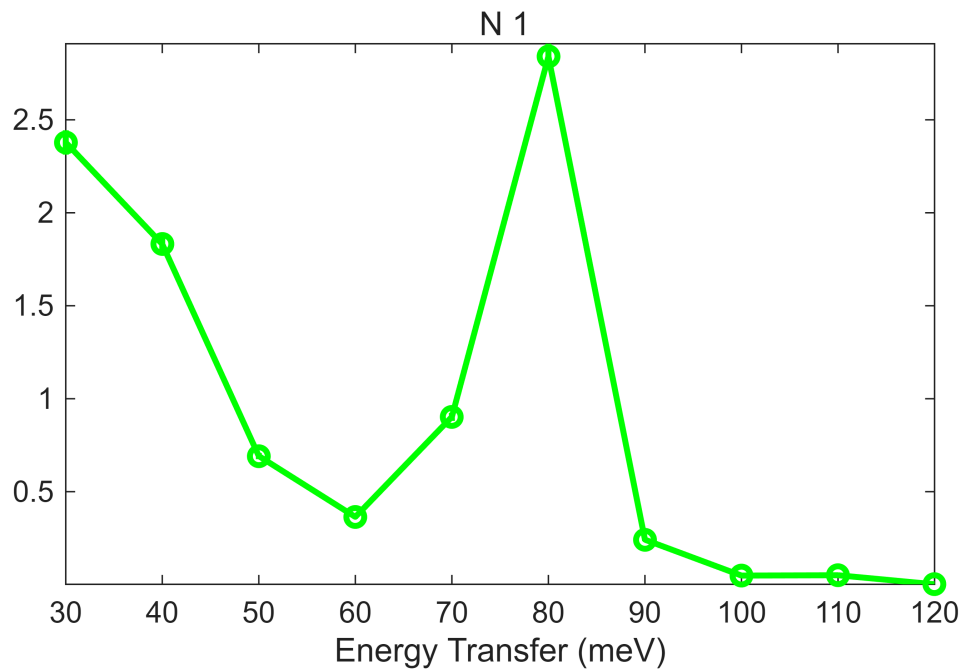
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Amplitude,
e:\SHARE\FE\Data\analysis\06_fit\with\J0sym4D_cutsAndFits\EnFit\Ei200off110_2D_dirGP_dE20_fitSlopeDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
J0,
e:\SHARE\FE\Data\analysis\06_fit\with\J0sym4D_cutsAndFits\EnFit\Ei200off110_2D_dirGP_dE20_fitSlopeDE.mat
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
Gamma,
e:\SHARE\FE\Data\analysis\06_fit\with\J0sym4D_cutsAndFits\EnFit\Ei200off110_2D_dirGP_dE20_fitSlopeDE.mat

beep
beep
beep

```

bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure;

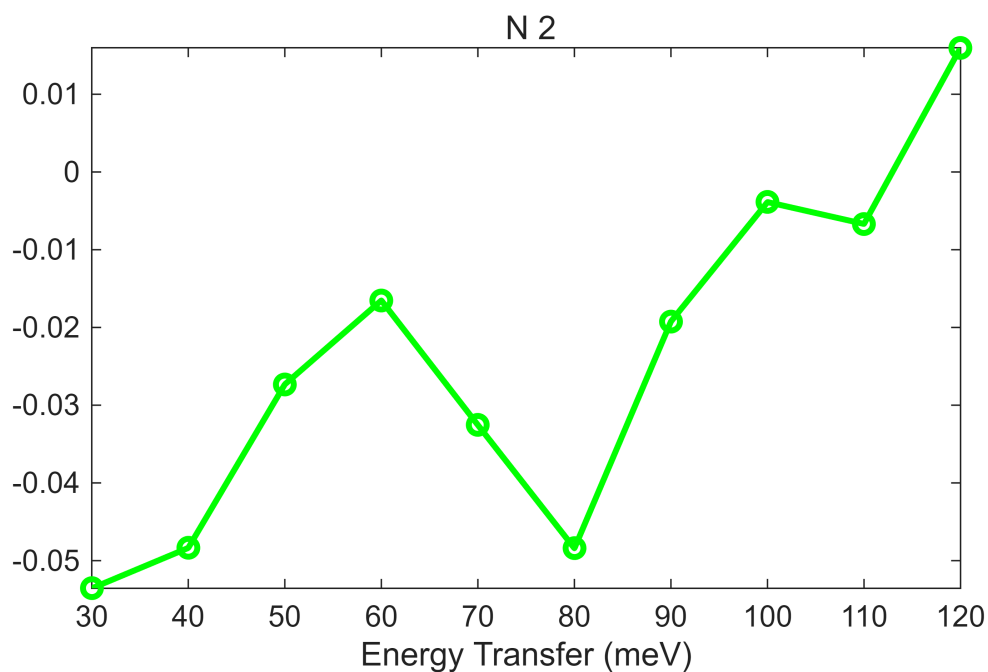
```



```

pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure

```



```

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure

```

```
fit_resEi200 = fit_res_rec;  
save(res_name, "fit_resEi200")
```